

Math 571 - Homework 2

Richard Ketchersid

Problems numbered [B:2.1:3] are from [Bergman's exercise notes](#). Problems labeled [R:2.1] are directly from Rudin. If I add an asterisk (*) to a problem, we have a slight variant of the problem.

Problem 2.1 (R:2.2*). A complex number γ is **algebraic** iff γ is a root to a polynomial with integer coefficients. Prove that there are complex numbers that are not algebraic.

The following are two fun facts just beyond what we got to in MAT-513:

Let $\mathbb{A} \subset \mathbb{C}$ be the set of algebraic numbers.

- (a) $\mathbb{A}$ is a field.
- (b) $\mathbb{A}$ is algebraically closed, that is, if α is a root of a polynomial in $\mathbb{A}[x]$, then $\alpha \in \mathbb{A}$. So in the definition of algebraic numbers you can use any ring of coefficients R , $\mathbb{Z} \subseteq R \subseteq \mathbb{A}$.

Problem 2.2 (B:2.2:4). This problem is concerning relative topology.

- (a) Suppose E is an open subset of a metric space X , and F is a subset of E . Show that F is open relative to E iff F is open.
- (b) Suppose E is a closed subset of a metric space X , and F is a subset of E . Show that F is closed relative to E iff F is open.
- (c) Suppose E is an open subset of a metric space X , and F is a subset of E that is closed relative to E . Show that F need not be open or closed in X .
- (d) Suppose E is a closed subset of a metric space X , and F is a subset of E that is open relative to E . Show that F need not be open or closed in X .

Problem 2.3 (B:2.2:20). Here you look at the Kuratowski game discussed in the podcast.

- (a) Show that $\text{Cl}(\text{Cl}(E)) = \text{Cl}(E)$ and $\text{Int}(\text{Int}(E)) = \text{Int}(E)$.
- (b) $\text{Int}(\text{Cl}(E)) = \text{Int}(\text{Cl}(\text{Int}(\text{Cl}(E))))$.
- (c) Show that you can get at most seven different sets from applying the operations of closure and interior repeatedly to a fixed set E .
- (d) Show that there is $E \subset \mathbb{R}$ so that all seven possible sets are realized.

Bonus: If you also throw in the operation of complement, then there are at most 14 possible sets you can get starting from a fixed set E .

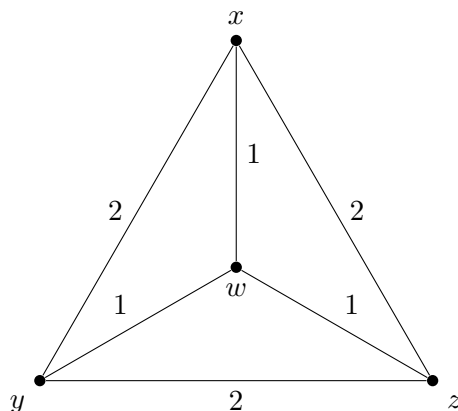
Problem 2.4 (B:2.2:14). This investigates the p -adic metric on $\mathbb{Z}$. Let $p > 1$ be an integer. Define $d_p(s, t) = p^{-e_p(s-t)}$ where $e_p(m) = k$ iff $p^k \mid m \wedge p^{k+1} \nmid m$, so $e_p(m)$ reports the maximal number of times m can be evenly divided by p . Note that $p^k \mid 0$ for all k , so it makes sense to define $e_p(0) = \infty$ and take $d_p(s, s) = p^{-\infty} = 0$.

Show that $d_p : \mathbb{Z} \times \mathbb{Z} \rightarrow [0, \infty)$ is a metric and in fact an **ultra-metric**, i.e., $d_p(q, s) \leq \max\{d_p(q, r), d_p(r, s)\}$.

For Fun: Show that all p -adic triangles are isosceles. This is a geometric idea; what corresponding topological fact is leading to this?

p -adics are very important and interesting number systems, particularly when p is a prime. [Here is a great video introduction to \$p\$ -adics](#), which discusses applications to solving Diophantine equations.

Problem 2.5 (B:2.2:18). Let X be a 4-element set. Consider a graph on the four nodes $\{x, y, z, w\}$ with edges and edge weights as shown.



- Verify that $d_X : X \times X \rightarrow [0, \infty)$ is a metric where $d(x, y) = d(x, z) = d(y, z) = 2$ and $d(w, x) = d(w, y) = d(w, z) = 1$.
- Show that there is no $F : X \rightarrow \mathbb{R}^k$ that is distance preserving, that is, $d_X(u, v) = d_{\mathbb{R}^k}(f(u), f(v))$.
- The example here is a 4-point space that cannot be *isomorphically* embedded in $\mathbb{R}^k$ for any k , yet every 3-point subspace can be isomorphically embedded in $\mathbb{R}^k$. Can you find a 5-point space that cannot be isomorphically embedded in $\mathbb{R}^k$ and yet all of whose 4-point subspaces can be isomorphically embedded in $\mathbb{R}^k$ for sufficiently large k .

Problem 2.6 (B:2.2:19). Let X be an infinite metric space. Show that there are uncountably many open subsets of X .

- Show that you can find $p_i \in X$ and $r_i > 0$ so that $N_{r_i}(p_i) \cap N_{r_j}(p_j) = \emptyset$.
- Associate with every $\alpha \in 2^{\mathbb{N}}$ the open set $O_\alpha = \bigcup_{\{i \mid \alpha(i)=1\}} N_{r_i}(p_i)$. Show that this map is an injection of $2^{\mathbb{N}}$ into the open sets of X .

- (c) (b) shows that there are at least continuum many open subsets of X , could the cardinality be even larger?

Problem 2.7 (R:2.26). Let X be a metric space in which every infinite set, A , has a limit point in A . Show that X is compact. (Follow the hints in Rudin.)

This is a very important property. We will see that for metric spaces, this gives the equivalence of " X compact" with " X is sequentially compact."

Problem 2.8 (B:2.3:3 - Compact sets get crowded!). Let X be a compact metric space and let $\epsilon > 0$. Show that there is an N so that for any $m \geq N$, if $p_1, \dots, p_m$ are m distinct points in X , then at least two of them are within ϵ of each other, that is, there are i, j with $d(p_i, p_j) < \epsilon$.

Problem 2.9 (B:2.5:1 - The usual definition of connected.). (a) Show that $E \subsetneq X$ is connected (as per Rudin) iff there is no non-empty proper subset of E that is **clopen** relative to E , i.e., closed and open relative to E .

(b) Show that $E \subsetneq X$ is connected (as per Rudin) iff there are two non-empty **relatively open subsets** of E that **cover** E .

Note: A space X is connected iff the clopen sets consist of exactly X and $\emptyset$.

Problem 2.10 (This is a mix of problems.). Consider the following collections of open subsets of $\mathbb{Q}$. (1) Verify that each gives a topology on $\mathbb{Q}$ and (2) compute for each, $\text{Cl}((0, \sqrt{2}))$, (3) show that the space is compact, or not, (4) show that the space is connected or not, and finally (5) how are these related to the standard relative topology that $\mathbb{Q}$ inherits from $\mathbb{R}$?

- (a) *The right-ray topology.* Take $\mathcal{O}$ to be the set of all right-rays of the form (a, ∞) for $a \in \mathbb{R} \cup \{+\infty, -\infty\}$.
- (b) *The co-finite topology.* Take $\mathcal{O}$ to be subsets of $\mathbb{Q}$ whose complements are finite together with $\emptyset$.
- (c) *The order topology.* A set A is open in this space iff for all $q \in A$, there is $q_1 < q < q_2$ in $\mathbb{Q}$ so that $(q_1, q_2) \cap \mathbb{Q} \subseteq A$.